Plastics Compounding Preliminary Product Review Meeting Outline

06 May 2004

Title:

Additives in Plastics Processing (Compounding) - Generic Scenario for Estimating Occupational Exposures and Environmental Releases

This document presents a standard approach for estimating worker exposures to and environmental releases of additives used in the compounding of plastic resins. It does not cover converting plastic resins into finished articles or the manufacture of polymers and additives.

Industry Sector Description:

The plastic manufacturing industry can be divided into four sections: polymer manufacturing, compounding, converting, and "in-house" manufacturing. This generic scenario will address compounding operations. Polymer manufacturing will not be included in this scenario. Compounders produce masterbatches of plastic resins with specific properties by blending the polymer (resin), additives, fillers, and reinforcements. Converters receive the masterbatch of plastic resin from compounders and form the compounded plastic resin into finished plastic products. A separate generic scenario will cover plastic resin converting into finish articles. Compounding and converting may take place at the same facility ("in-house" manufacturing) or separate facilities. However, in-house manufacturing will not be covered in this generic scenario. As a conservative estimate it is assumed that compounding and converting will take place at separate sites.

There are two types of polymers:

- *Thermoplastic polymers* are melted and become fluid when heat and pressure are applied. The molten polymers are formed into a shape via pressure. Thermoplastics solidify when cooled and the heating and cooling process can be repeated many times with little loss in properties. This generic scenario focuses on thermoplastic polymers.

- *Thermosetting polymers* (e.g., foam, epoxy) are formed into finishing products during a chemical reaction under pressure and heat. This process creates permanent cross-linking and the product retains it shape during subsequent cooling and heating. Thermosetting polymers are not within the scope of this generic scenario and are not discussed further.

NAICS code 325991, custom compounding of purchased resins, comprises establishments primarily engaged in custom mixing and blending of plastic resins made elsewhere or reformulating plastic resins from recycled plastic products. According to the 2001 Census Bureau County Business Patterns, there are 715 establishments and 25,557 workers in NAICS code 325911. This NAICS code represents the compounding industry subsection discussed in this scenario. Table 1 presents the types of thermoplastic resins, common uses, and 2003 production volume.

Table 1: Type of Thermoplastics, North American Production, And Common Uses

Resin	Production		Common Uses
	millions of pounds, dry weight basis	millions of kg, dry weight basis	
Low density polyethylene (LDPE) (a)	7,846	3,556	Packaging films, plastic bags, molded housewares, toys, containers, pipes, drums, gasoline tanks, coatings
Linear low density polyethylene (LLDPE) (a)	11,174	5,079	
High density polyethylene (HDPE) (a)	15,691	7,132	
Polypropylene (PP) (a)	17,559	7,981	Packaging, automotive, appliances, carpeting
Acrylonitrile butadiene (ABS) (b)	1,215	552	Appliances, automotive parts, pipe, business machines, telephone components
Nylon (b)	1,282	583	Automotive parts, electronics, packaging
Polystyrene (a)	6,442	2,928	Packaging, food service products, automotive parts, toys, housewares
Styrene Acrylonitrile (SAN) (b)	137	62	Automotive instrument panels and interior trim, housewares
Other sterics (b)	1,597	726	NA
Polyvinyl Chloride (PVC) (a)	14,696	6,680	Upholstery, flooring, wall coverings, pipe, siding, apparel
Thermoplastic polyester (b)	7,587	3,449	Plastic bottles, textiles, X-ray film, magnetic tape, packaging, metallized film, strapping, labels
Total Selected Thermoplastics	**85,226**	**38,728**	

Sources: The American Plastics Council and The Society of Plastic Engineering.
(a) Canadian production included.
(b) Canadian and Mexican production included.
NA – Not available.

Sources:

OECD 2003. Emission Scenario Document on Plastic Additives. Organisation for Economic Co-operation and Development. OECD Environmental Health and Safety Publications. July 2003.

Kirk-Othmer Encyclopedia of Chemical Technology. Plastics Processing, Plasticizers, Flame Retardants. 4th ed., 1991.

U.S. Census. County Business Patterns. U.S. Census Bureau. 2001. http://censtats.census.gov/cbpnaic/cbpnaic.shtml (last confirmed December 2003)

The American Plastics Council 2003. APC Year End Statistics for 2003. www.americanplasticscouncil.org/benefits/economics/economics.html (last confirmed March 2004)

The Society of Plastics Industry. 2003. Definitions of Plastic Resins. www.plasticsindustry.org/industry/defs.htm (last confirmed February 2003)

Process Description:

Polymer resin is received from the resin manufacturer in the form of pellets. A compounding site blends the resin and additives to produce a masterbatch. The converting site then processes the masterbatch by shaping the plastic into the desired form for the final plastic product. Converting operations are not covered in this generic scenario, refer to *Additives in Plastics Processing (Converting into Finished Products)* generic scenario for information on converting operations. Compounding and converting may take place at the same facility ("in house" manufacturing) or separate facilities. This scenario estimates that the compounding of the plastic resin and the converting of the resin into plastic products take place at separate facilities; therefore, "in house" manufacturing is not covered in this scenario.

Additives are typically added to polymer resins to adjust the properties of the plastic resin. Additive components include antioxidants, antistatics, blowing agents, colorants, coupling agents, fillers, flame retardants, heat stabilizers, impact modifiers, lubricants, plasticizers, preservatives, slip promoters, and ultraviolet stabilizers. A search of CEB's PMN database resulted in almost 70 plastic additive cases, including most of the additive types mentioned above. Table 2 presents the additive type, function, types of chemicals, physical state, and default weight fraction of additive in plastic resin. Table 3 presents typical weight fractions of additives in specific types of plastics resins.

Table 2. Types of Additives Used in Plastic Processing

Additive Type	Function	Types of Chemicals	Typical Physical State	Default Fraction In Plastic Resin
Antioxidants	Inhibit the oxidation of plastic materials that are exposed to oxygen or air at normal or high temperatures.	Alkylated phenols, amines, organic phosphites and phosphates, esters	Solid powder or pellets (a)	0.005 (a)

Additive Type	Function	Types of Chemicals	Typical Physical State	Default Fraction In Plastic Resin
Antistatics	Impart a minimal to moderate degree of electrical conductivity to the plastic compound, preventing electrostatic charge accumulation on the finished product.	Quaternary ammonium compounds, anionics, amines	Solid (a)	0.01 (a)
Blowing agents	Releases gas during heating to produce a cellular form of plastic (typically used in thermosetting resins) (a)	Carbon dioxide, acetone, sodium bicarbonate, organic nitrogen compounds (a)	Solids, volatile liquids, compressed liquids, gases (a)	0.06 (a)
Colorants	Impart color to the plastic resin.	Titanium dioxides, iron oxides, anthraquinones, carbon black	Fine powder (pigments) and liquid or waxy solid (dyes) (a)	0.2 (a)
Coupling Agent	Interface between filler and plastic, bonding with both phases to improve interfacial adhesion. Typically introduced during the treatment stage of filler manufacture. (a)	Organometallic compounds, silanes (a)	Low viscosity liquid or low melting point solid (a)	0.03 (a)
Curing agent	Assist in the curing of thermosetting materials (a)	Peroxides, amines, organotin compounds (a)	Solids and liquids	0.03 (a)
Fillers	Inert materials which reduce polymer cost, improve processability, and improve mechanical properties	NA	Solid (fiber or powder)	0.55 (a)
Flame Retardants	Reduce the tendency of the plastic product to burn.	Antimony trioxide, chlorinated paraffins, bromophenols	Solid (a)	0.4 (a)
Heat Stabilizers	Assist in maintaining the chemical and physical properties of the plastic by protecting it from the effects of heat such as color changes, undesirable surface changes, and decreases in electrical and mechanical properties.	Lead, barium-cadmium, tin, calcium zinc	Solid (sometimes liquid) (a	0.05 (a)
Impact Modifiers	Prevent brittleness and increase the resistance of the plastic to cracking.	Natural rubber, acrylonitrile and ethylene as copolymers	Granular solid (a)	NA
Lubricants	Assist in easing the flow of the plastic in the molding and extruding processes by lubricating the metal surfaces that come into contact with the plastic.	Stearic acid, waxes, fatty acid esters, fatty acid amines	Waxy solid; soft powder (a)	0.012 (a)
Plasticizers	Increase the plastic product’s flexibility and workability.	Adipates, azelates, trimellitates, DOP/DIOP/DIDP	Liquids or waxy solids (i.e., low melting point) (a)	0.5 (a)
Preservatives	Protects against fungi and bacteria	Organotin, organomercury compounds (a)	Solid (a)	NA

Additive Type	Function	Types of Chemicals	Typical Physical State	Default Fraction In Plastic Resin
Slip promoters	Improve surface lubrication during processing and use (a)	Ca and Zn stearates, waxes and fatty acid amines or esters (a)	Soft powders or waxy solids (a)	0.25 (a)
Ultraviolet Stabilizers	Absorb or screen out ultra-violet radiation thereby preventing the premature degradation of the plastic product.	Benzophenones, benzotriazole, salicates	Solids (a)	0.05 (a)

Source: Kirk-Othmer

(a) Information was obtained from Emission Scenario Document on Plastic Additives. July 2003. High end of range of weight fraction is presented as default value for more specific information see Table 3.

Table 3. Fraction of Plastic Additives in Plastics Resins

	Typical Fraction of Additive in Each Type of Plastic												
Additive	**LBDE**	**HBPE**	**Poly-propylene**	**Rigid PVC**	**Flexible PVC**	**Poly-styrene**	**Expanded Poly-styrene**	**ABS**	**PET**	**Poly-amides**	**Acrylics**	**Acetals**	**Poly-carbonate**
Antioxidant	0.001	0.0025	0.005	0.002	-	-	-	-	-	-	-	-	-
Antistatic	0.001	0.003	-	-	-	-	-	-	-	-	-	0.01	-
Blowing agent	0.04	-	-	0.04	-	-	0.06	-	-	-	-	-	0.04
Colorants	0.03	0.03	0.03	0.005-0.03	0.02	0.01	-	0.05	0.03	0.2	0.01	0.02	0.03
Coupling agents	-	-	-	-	-	-	-	-	-	0.005	0.005	0.005	0.005
Curing agent	0.03	-	0.005	-	-	-	-	-	-	-	-	-	-
Fillers	-	-	0.4	0.1	0.1-0.5	-	-		0.55	0.4	0.35	0.3	0.4
Flame Retardant	0.2	0.2	0.04-0.4	-	0.05	0.2	0.15	0.2	0.15	0.15-0.25	-	-	-
Heat Stabilizers	-	-	-	0.01-0.05	0.02-0.05	-	-	0.0075-0.01	-	0.005	-	0.002	0.0015-0.0025
Lubricants	-	-	-	0.01	0.001-0.003	0.012	-	-	-	-	-	-	0.012
Plasticizers	-	-	-	-	0.3-0.5	-	-		0.05	0.01			
Slip promoters	0.0005	-	-	-	-	-	-	-	-	-	-	0.25	-
UV Stabilizers	-	-	0.05	0.005	-	0.005	-	0.005	0.005	0.03	0.002	0.03	0.003

Source: Emission Scenario Document on Plastic Additives. July 2003.
LBDE – Low density polyethylene
HPDE – High density polyethylene
PVC – Polyvinyl chloride
ABS – Acrylonitrile butadiene
PET – Polyethene terephthalate

There are several methods used to perform the blending of masterbatches.

1. *Tumble Blenders:* Used for preblending solids of similar particle size. Polymer powders and dry additives are charged to a closed container that is rotated/tumbled.

2. *Ball Blenders:* Used for dry or liquid polymer systems. Similar to tumbling with mills to enhance fine dispersion.

3. *Gravity Mixer:* Used for dry or liquid polymer systems. Materials are cascaded downwards through a series of baffles and repeated passes are possible with an elevator or auger.

4. *Paddle/Double Arm Mixer:* Used for high viscosity mastics, pastes, elastomers, dough molding compounds, and materials with high filler content. Contra-rotating blades knead and fold materials.

5. *Intensive Vortex Action Mixers:* Used for dry or liquid polymer systems. Materials are preheated and mixed with high-speed stirrers.

6. *Banbury Internal Mixers:* Used for dry or liquid polymer systems. A ram presses the polymer and additives against revolving rotor blades in an enclosed chamber. Temperature control is necessary to avoid degradation of materials. The resin is typically sent to secondary mixing or extrusion processes.

7. *Two Roll Mill:* Used for dry or liquid polymer systems. Two rollers contra-rotate at different speeds that generate a shearing action. Polymer and additives are introduced to the rollers and the rollers knead the materials.

8. *Extruders:* Polymer and additives are preblended in a hopper or tumbler then the mixture is fed into an extruder comprised of one or two screws. The material is heated in the extruder and the screws shear the material and transport it through the extruder.

Tumble blenders, ball blenders, gravity mixers, paddle/double arm mixers, intensive vortex action mixers, and banbury internal mixers are all closed systems. Two roll mills and extruders are partially open systems. The compounding process may consist of one or more of the above described processes. Typically the blending process (methods 1 through 6) are followed by a secondary mixing or extrusion process. For each of the blending processes described, the additive materials are unloaded from its transport container and charged to the mixer (Release 1, Exposure A). For mixing of dry components (additives and resin), particulate emissions are expected from transferring and mixing activities (Release 2). Most plastic additives are non-volatile at ambient temperatures; however, for processes that use or create heat (e.g., extrusion, two roll mill, banbury internal mixing, and intensive vortex mixer) volatile emissions are expected (Release 3). Equipment cleaning as well as cooling water waste is expected to take place at compounding sites (Release 4, Exposure B). However, specific details on frequency and estimates on losses are not currently available. The masterbatch is loaded into the transport container (Exposure C) and sent to the converter in pellet, sheet, film, or pipe form.

Sources:

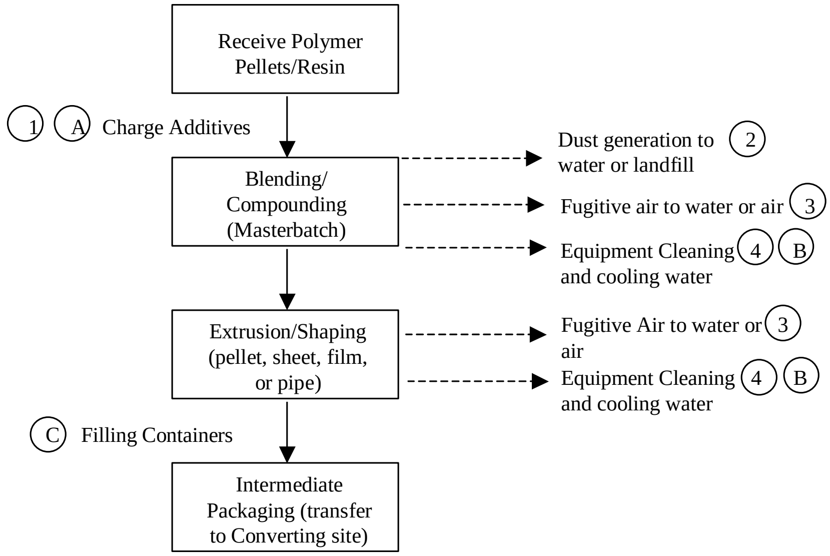
OECD 2003. Emission Scenario Document on Plastic Additives. Organisation for Economic Co-operation and Development. OECD Environmental Health and Safety Publications. July 2003.

Kirk-Othmer Encyclopedia of Chemical Technology. Plastics Processing, Plasticizers, Flame Retardants. 4th ed., 1991.

U.S. EPA. Profile of the Rubber and Plastics Industry. U.S. EPA Office of Compliance Sector Notebook. EPA 310-R-95-016. September 1995.

EPA 1984. Development Document for Effluent Limitations Guidelines and Standards for the Plastics Molding and Forming Point Source Category. Final Report. December 1984. EPA/440/1-84/069

Process Diagram:



<u>Environmental Releases:</u>

1. Container residue from additive transport container released to water, incineration, or landfill.
2. Dust generation from transferring/compounding released to water or landfill.
3. Fugitive air emissions from compounding/shaping released to water or air.
4. Equipment cleaning and cooling water from compounding released to water.

<u>Occupational Exposures:</u>

A. Exposure to solid or liquids during unloading of additives from transport container and charging additives to operation.
B. Exposure to liquids during equipment cleaning of compounding equipment.
C. Exposure to solids during filling containers with compounded plastic resin.

General Facility Estimates:

I. Days of Operation

CEB standard assumption, 250 days per year, based on five day work week and two weeks per year of operation shut down.

II. Daily Use Rate of Plastic Additive

Estimate based on the total amount of thermoplastics manufactured in the U.S. (The American Plastics Council 2002, see Table 1), the total number of plastic compounding sites (715 establishments, 2001 County Business Patterns), days of operation (CEB standard assumption), and the fraction of the additive in the thermoplastic (see Tables 2 or 3). This throughput will be adjusted by the percent of the chemical of interest in the additive if the additive is a mixture of chemicals. If the type of plastic is not known, the total amount of plastic resin produced in the U.S. should be used from Table 1 (35,012 million kg/yr). This estimate assumes that all compounding sites (715 establishments) produce all kinds of plastic resins.

Daily use rate = amount of resin / (# compounding sites x days of operation) x fraction of additive x fraction of chemical in additive

Example of antioxidant used in polyester plastic resin:
96 kg/site-day = 3,449,000,000 kg/yr of polyester resin / 715 sites / 250 days/yr x 0.005 x 1

III. Number of Compounding Sites

Estimate based on the total amount of plastic additive produced (PV, kg/yr); the daily use rate of plastic additive; and the days of operation.

compounding sites = PV / (daily use rate of additives x days of operation)

Example of antioxidant (0.005 weight fraction, Table 2) for an assumed production volume of 100,000 kg/yr:
4 sites = 100,000 kg/yr / (96 kg/site-day x 250 days/yr)

If the number of sites is less than one, recalculate the days of operation.

days of operation = PV / daily use rate of additives

The values for days of operation, daily use rate of plastic additive, and number of compounding sites are related. This scenario presents an equation to calculate the daily use rate of the plastic additive from the amount of resin produced, total number of sites, and days of operation with default values for each. The daily use rate and the supplied production volume are then used to calculate the number of compounding sites.

If the number of compounding sites is known, the daily use rate can be calculated directly without the use of the equation in section II. This alternative calculation is: daily use rate = production volume / number of sites. However, it is recommended to calculate the daily use rate based on information in section II and compare it to the daily use rate based on number of sites.

Environmental Releases

1. <u>Container Residue from Original Additive Transport Container (Release 1)</u>

   Most additives are received as solids in bags. For additives received as solids, 1% loss is estimated (OECD 2003). This estimate is consistent with CEB standard assessments for solid residual in containers. For additives received as liquids, 3% loss is estimated (CEB standard assessment). The media of release is uncertain; therefore, container residue is assumed to be released to water, landfill, or incineration (CEB standard estimate). Specific information on size of containers is not available; therefore, 55 gal drum (208 L) is assumed as the default. 1 kg/L is assumed as the default density of the additive if specific information is not available. The information presented for this standard CEB model/estimate is based on the current version of the model (as of the date of this generic scenario). Standard CEB models are subject to change; therefore, the current version of the standard CEB model should be used.

   Number of Containers (containers/site-year) = Daily Use Rate (kg/site-day) x Days of Operation (days) / Fraction of Chemical in additive / Container Volume (L/container) / Density (kg/L)

   Example for antioxidant transported in drums:
   115 containers/site-yr = 96 kg/site-day x 250 days / (1 x 208 L/container x 1 kg/L)

   If the number of containers is less than the days of operation, the days of release is equal to the number of containers, and the daily release is calculated based on the following formula:

   Daily Release from Container Residue (kg/site-day) = Container Volume (L/container) x Density (kg/L) x Fraction of Chemical in additive x Loss Fraction

   Example for solid antioxidant transported in drums:
   2.08 kg/site-day = 208 L/container x 1 kg/L x 1 x 0.01

   If the number of containers is greater than the days of operation, the days of release is equal to the days of operation, and the daily release is calculated based on the following formula:

Daily Release from Container Residue (kg/site-day) = Daily Use Rate (kg/site-day) x Loss Fraction

2. <u>Dust from Compounding Released to Water or Landfill (Release 2)</u>

Solid Additives: Dust generation is expected from transferring and mixing activities of solids (0.65% loss of daily use rate (range of 0.65% for fine particles (particle size <40μm) and 0.21% for course particles (particle size > 40μm) based on estimates for filler additives), OECD 2003). Particles are originally released to air but are expected to eventually settle to the ground. Dust particles are expected to be cleaned from the equipment and floor with water or disposed of directly to landfill.

Daily release from dust = daily use rate x loss fraction

Example of solid antioxidant additive:
0.62 kg/site-day released = 96 kg/site-day x 0.0065 loss fraction

3. Fugitive Air from Compounding Released to Water or Air (Release 3)

Solid and Liquid Additives: Most plastic additives are non-volatile; however, mixing operation may be performed at elevated temperatures; thus slight volatilization may occur. For solid and liquid additives, 0.05% loss of daily use rate is estimated (OECD 2003, loss rates are based on high (0.05%), medium (0.01%), and low (0.002%) volatility at 200°C for typical plasticizers). Particles are originally released to air but subsequent condensation may result in losses to water. As a conservative estimate 50% is assume to be released to water and 50% is assumed to be released to the atmosphere (OECD 2003).

Daily release to water from volatilization = daily use rate x loss fraction

Example of solid antioxidant additive:
0.024 kg/site-day released = 96 kg/site-day x 0.00025

Daily release to air from volatilization = daily use rate x loss fraction

Example of solid antioxidant additive:
0.024 kg/site-day released = 96 kg/site-day x 0.00025

The table below shows the vapor pressures of common plasticizers and volatility groupings used in the OECD document.

Temperature (°C)	Vapor Pressure (hPa; 1 hPa = 0.75 mm Hg			
	DEHA	**BBP**	**DEHP**	**DIDP**
20	6.3×10^{-7}	6.5×10^{-6}	2.2×10^{-7}	3.3×10^{-8}
50	2.7×10^{-5}	1.5×10^{-4}	1.0×10^{-5}	1.8×10^{-6}
100	3.7×10^{-3}	9.1×10^{-3}	1.6×10^{-3}	3.4×10^{-4}
150	0.16	0.21	0.076	0.019
200	3.1	2.5	1.6	0.46
Volatility grouping	high	high	medium	low

4. Residual from Compounding Equipment Cleaning (Release 4)

This release includes cooling water used during compounding and equipment cleaning water. The release is based on CEB standard estimate for equipment cleaning of entire process (2% loss of daily use rate) released to water. (Sector Notebook and EPA 1984 state that water is used as a direct coolant routinely during extrusion and compounding). The use of cooling water and water to clean equipment is a conservative estimate. The Development Document for Effluent Limitation Guidelines for the Plastics Molding and Forming Point Source Category (1984) estimates that 1,898 sites use process water (this includes converting processes). Process water includes contact cooling and heating water, cleaning water, and finishing water. Although, this information is specific to converting processes, as a conservative estimate it is reasonable to assume that compounding sites may use water.

Daily release from equipment cleaning = daily use rate x loss fraction

Example for antioxidant additive
1.9 kg/site-day released = 96 kg/site-day x 0.02

<u>Control Technologies:</u>

Water: According to the Development Document for Effluent Limitation Guidelines for the Plastics Molding and Forming Point Source Category (1984), approximately 31% of surveyed sites directly discharged their process water; 44% indirectly discharged (POTW); and 25% had a zero discharge. Zero discharge methods include recycling, evaporation pond, septic tank with leach field, evaporation from equipment, land application, and contract haul. Types of on-site treatment include settling, pH adjustment, activated sludge, activated carbon adsorption, filtration, and vacuum filtration.

Air: The Emissions Scenario Document on Plastic Additives suggests that bag filters used to collect particulate emissions are 99% efficient. However, the prevalence of bag filter use was not available.

Occupational Exposures:

Number of Workers Exposed per Site

Estimate based on total number of workers in NAICS code 325991 (25,557 workers, County Business Patterns), total number of sites (715 sites, County Business Patterns), and percent of production workers in industry (66%, Annual Survey of Manufacturers).

workers/site exposed = total # workers in industry / number of sites x % of production workers

24 workers/site exposed = 25,557 workers in industry / 715 sites x 0.66

A. Exposure from Unloading and Charging Additives to Compounding Process (Exposure A)

Solid or liquid additives will be transferred manually or automatically from transport containers to mixing equipment. Workers are expected have dermal exposure to liquids and inhalation and dermal exposure to solids from transferring activities. Specific information on the number workers involved in this activity is not available; therefore, it is assumed that at least 8 workers will be involved in unloading and charging operations. Information regarding personal protective equipment (PPE) and engineering controls for this activity was not available.

Inhalation: For solids, CEB standard estimate based on OSHA PEL for particulates not otherwise regulated (daily use rate > 54 kg/site-day) or small dye weighing scenario (daily use rate < 54 kg/site-day). It most case it is expected that more than 54 kg/site-day of a solid additive will be handled. For liquids, inhalation exposure is negligible (VP < 0.001 torr).

Inhalation exposure = OSHA PEL x breathing rate x hours x fraction of chemical in additive (if applicable)

Example for solid additive:
150 mg/day exposure = 15 mg/m^3 x 1.25 m^3/hr x 8 hrs x 1

Dermal: CEB standard estimate for routine direct handling of solids, 2 hands and incidental contact of liquids, 2 hands.

Solid exposure:
Dermal exposure = 3,100 mg/day x fraction of chemical in additive (if applicable)

Example for solid additive:
3,100 mg/day exposure = 3,100 mg/day x 1

Liquid exposure:
Dermal Exposure = liquid amount on skin x area of skin x # of incidents x fraction of chemical in additive

Example of liquid additive:
1,800 mg/day exposure = 2.1 mg/cm^2-incident x 840 cm^2 x 1 incident/day x 1

B. <u>Exposure from Compounding Equipment Cleaning (Exposure B)</u>

Workers are expected to be exposed to plastic additives during manual equipment cleaning operations. Specific information on equipment cleaning was not available; therefore, it is assumed that equipment cleaning may take play manually. Dermal exposure is expected from manual equipment cleaning. Specific information on the number workers involved in this activity is not available; therefore, it is assumed that at least 8 workers will be involved in equipment cleaning activities. Information regarding PPE and engineering controls for this activity was not available.

Inhalation: Not expected, particles are expected to be contained in water.

Dermal: Assume manual equipment cleaning dermal model. CEB standard estimate for routine contact of liquids, 2 hands for equipment cleaning waste. Assume concentration of chemical in cleaning solution is 50% the concentration of chemical in resin (engineering judgment). Note, dermal contact with cooling water is not expected because of elevated temperatures.

Dermal Exposure = liquid amount on skin x area of skin x # of incidents x fraction of chemical in additive x fraction of additive in resin x fraction of resin in water

Example of antioxidant in resin at 0.005 weight fraction:
4.4 mg/day exposure = 2.1 mg/cm^2-incident x 840 cm^2 x 1 incident/day x 1 x 0.005 x 0.5

C. <u>Exposure from Filling Containers with Compounded Resin (Exposure C)</u>

Plastics are transported as pellets, sheets, films or pipes (Kirk-Othmer), which consist of large particles sizes. Dust generation is not expected during packaging; therefore, inhalation is negligible. Additive has been incorporated into plastic material, some surface contact may occur; however, dermal exposure is nonquantifiable (CEB Method for Screening-Level Assessments of Dermal Exposure). Specific information on the number workers involved in this activity is not available; therefore, it is assumed that at least 8 workers will be involved in filling operations. Information regarding PPE and engineering controls for this activity was not available.

Other Sources:

Department of the Environment UK. Use Category Document: Plastic Additives. Prepared by RAPRA Technology Limited and the Building Research Establishment. June 1998.

Annual Survey of Manufacturers. Statistics for Industry Groups and Industries: 2001. U.S. Census Bureau. January 2003.

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NIOSH 1998. Health Hazard Evaluation Report 97-0154-2693. Siemens Energy & Automation, Inc. Urbana, OH. June 1998.

U.S. EPA OSW 2003. Municipal Solid Waste in the United States: 2001 Facts and Figures. Office of Solid Waste and Emergency Response. EPA530-R-03-011. October 2003.

U.S. Census. County Business Patterns. U.S. Census Bureau. 2001. http://censtats.census.gov/cbpnaic/cbpnaic.shtml (last confirmed December 2003)

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Radian 1984. Industrial Process Profiles for Environmental Use, Chapter 10, Plastics and Resins Production Industry, Radian Corporation, 1984.

EPA 1984. Development Document for Effluent Limitations Guidelines and Standards for the Plastics Molding and Forming Point Source Category. Final Report. December 1984. EPA/440/1-84/069

AP-42 Analogous Industries:

ERG reviewed the AP-42 information for data from analogous industries. Related industry data identified includes: Plastic resin manufacturing for 1) PVC, 2) poly(ethylene Terephthalate), 3) polystyrene, and 4) polypropylene. PVC, Poly(ethylene terephthalate), and polypropylene AP-42 sections both present particulate emission factors. Although, this generic scenario does not include polymer manufacturing in the scope, the process description for the manufacture of poly(ethylene terephthalate) and polypropylene polymers include a chopper/crushing operation. This process may be similar to mixing operations for compounding. Polystyrene AP-42 section does not present particulate emission factors but does indicate in the process description an extrusion and pelletizer operation. AP-42 section on polyester resin plastic products fabrication has been removed from the EPA website stating that emission factors appear to under predict styrene emissions. The table below summaries the available emission factors for related industries.

Emission Factors for Plastic Manufacturing Operations from AP-42

Type of plastic/operation	Particulate		Gases	
	Emission factor	Notes	Emission factor	Notes
PVC	17.5 kg/Mg	Usually controlled with fabric filter, efficiency 98-99%	8.5 kg/Mg	as vinyl chloride
Poly(ethylene terephthalate)				
Product storage	0.0003 g/kg	0.4 g/kg if no controls	NA	
Total plant	0.17 g/kg		NA	
Polystyrene (extrusion quench vent)	NA		0.01-0.3 VOC/kg	non-methane VOC
Polypropylene	1.5 kg/Mg		0.35 kg/Mg	as propylene

Below is a comparison of PVC particulate emission factor (highest factor) with releases calculated for Releases 2 and 3.

AP-42 Emission factor for colorant additive in PVC manufacturing:
6,680,000,000 kg/yr of PVC / 715 sites / 250 days/yr x 1 Mg/1000 kg x 17.5 kg/Mg x 0.02 (fraction of additive) = *13 kg/site-day released to air*

OECD release factor of 0.007 (combination of Release 2 and 3):
6,680,000,000 kg/yr of PVC / 715 sites / 250 days/yr x 0.02 (fraction of additive) x 0.007 = *5.2 kg/site-day released to air*

The highest emission factor from an analogous industry (manufacture of PVC) is within an order of magnitude of the OECD based estimates. This shows that estimates based on OECD data is reasonable.